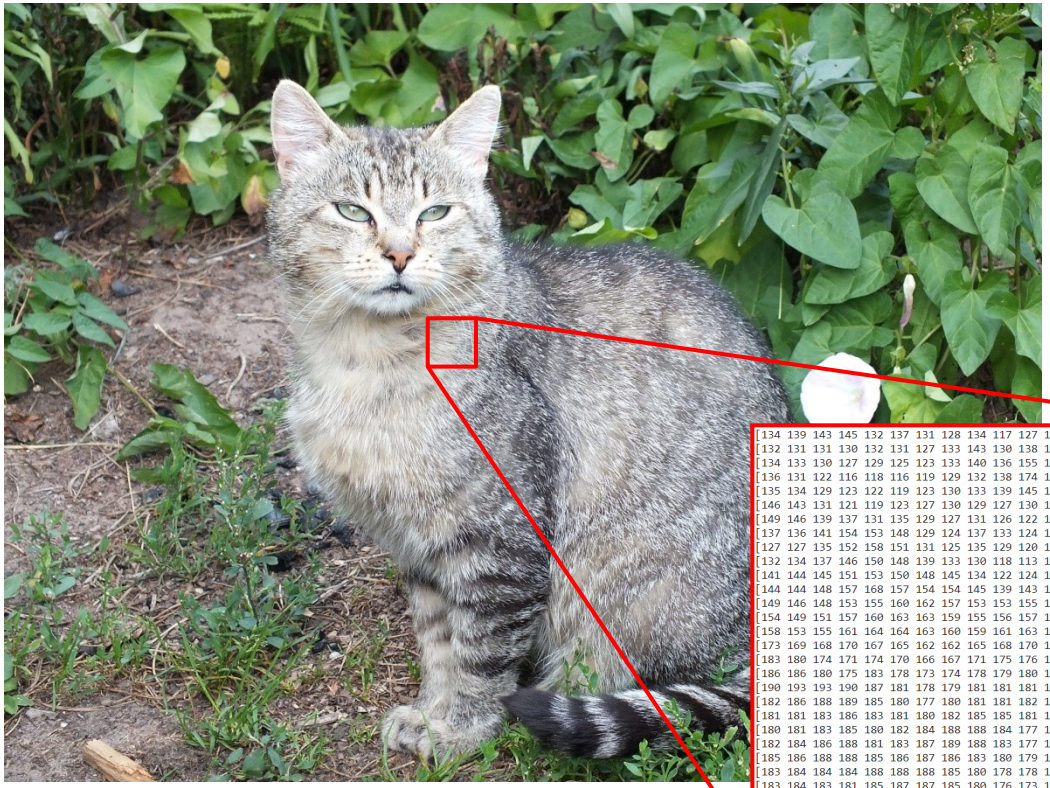
Computer Vision with Embedded Machine Learning

What is Image Classification?

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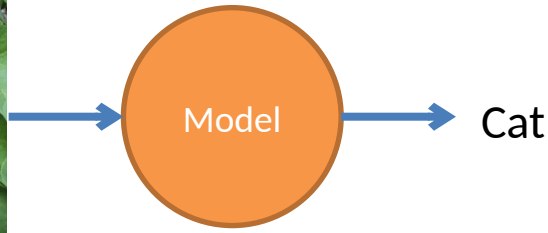


Resolution: 2048 x 1536 pixels
Bit depth: 24 bits

Numpy array shape: (1536, 2048, 3)

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Binary Classification

One-vs-One



vs.



One-vs-Rest



vs.



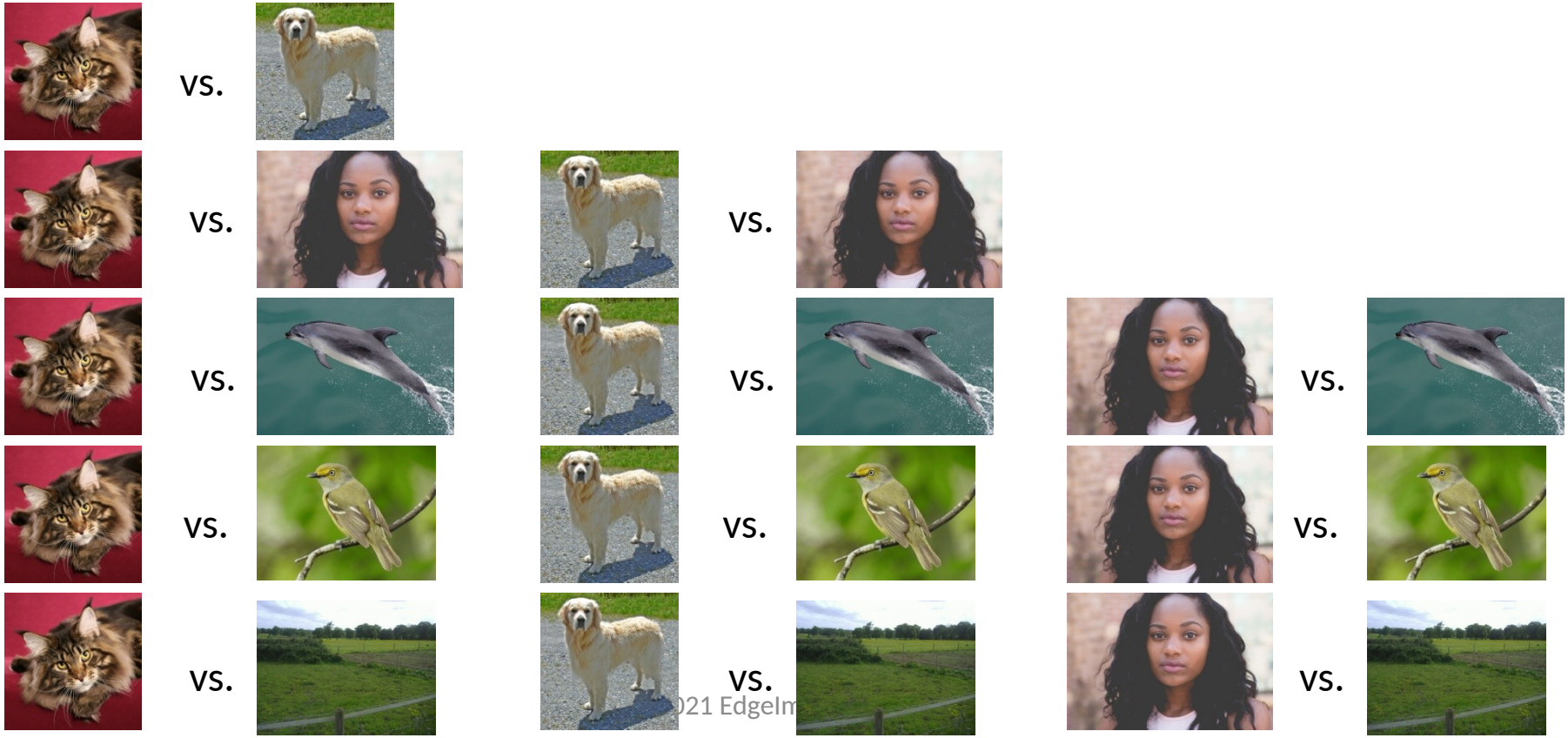
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Multiclass Classification

 vs.  vs.  vs.  vs.  vs. 

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Multiclass: One-vs-One



021 Edgelm

Multiclass: One-vs-Rest



vs.



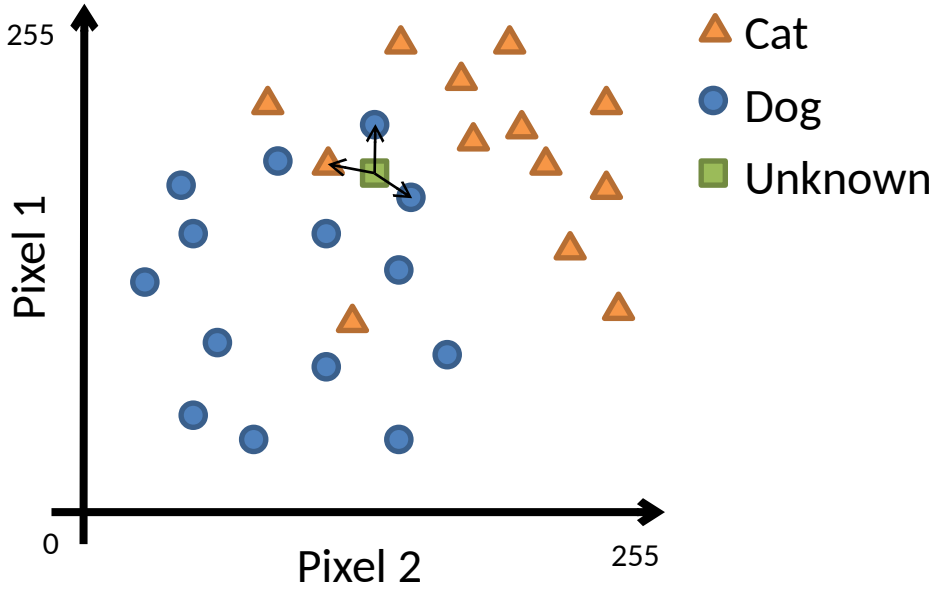


vs.



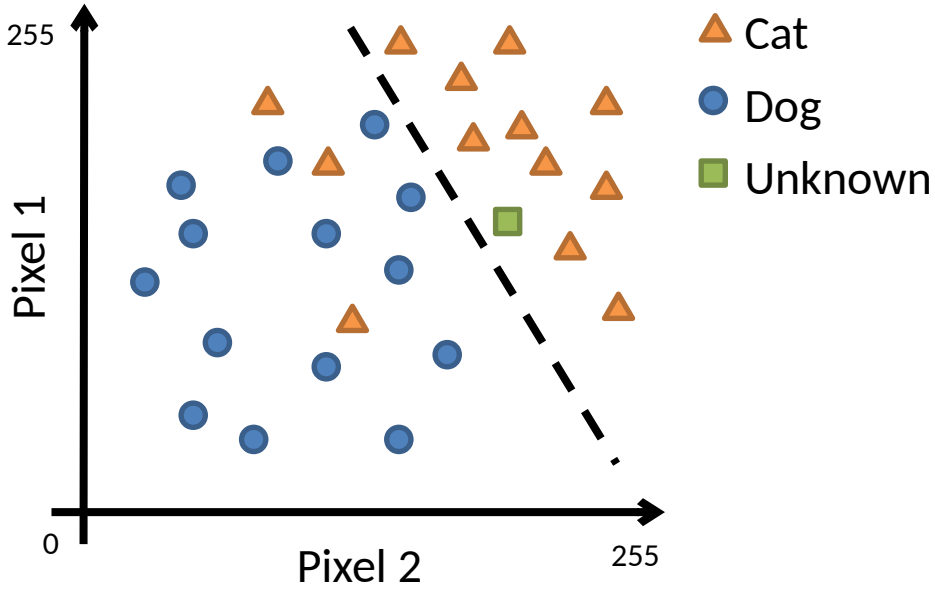
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k-Nearest Neighbors (k-NN)



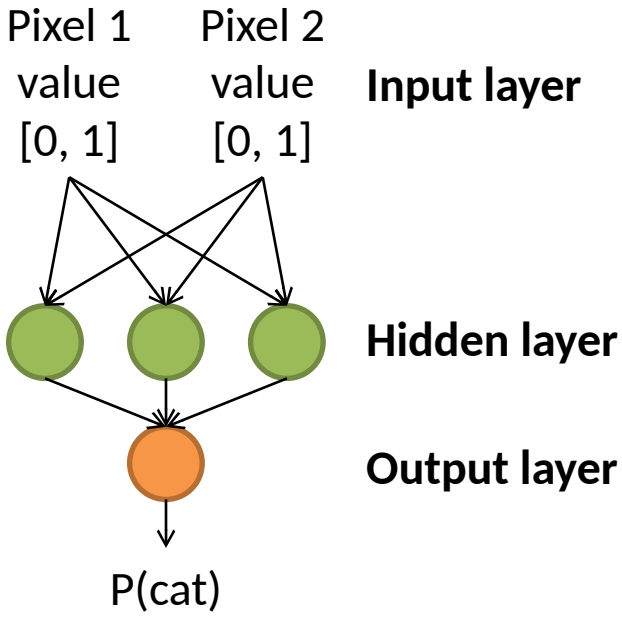
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Support Vector Machine (SVM)



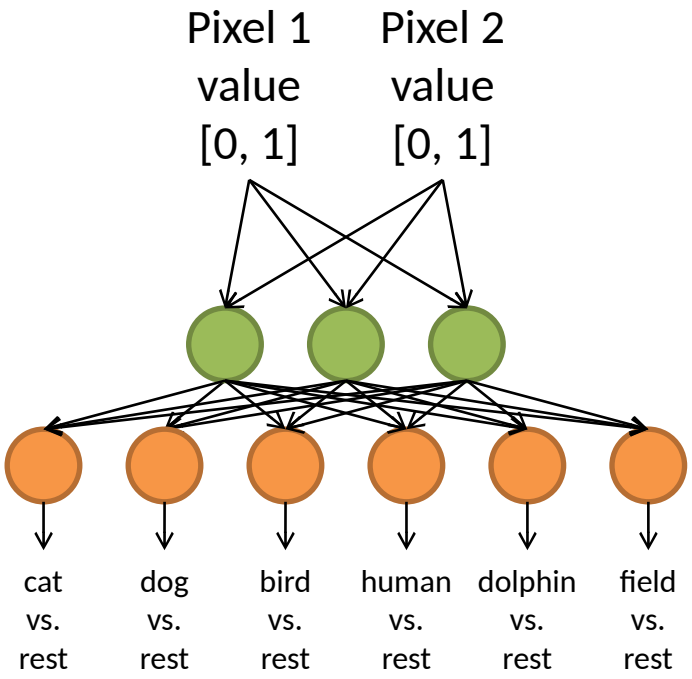
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Neural Network (NN)



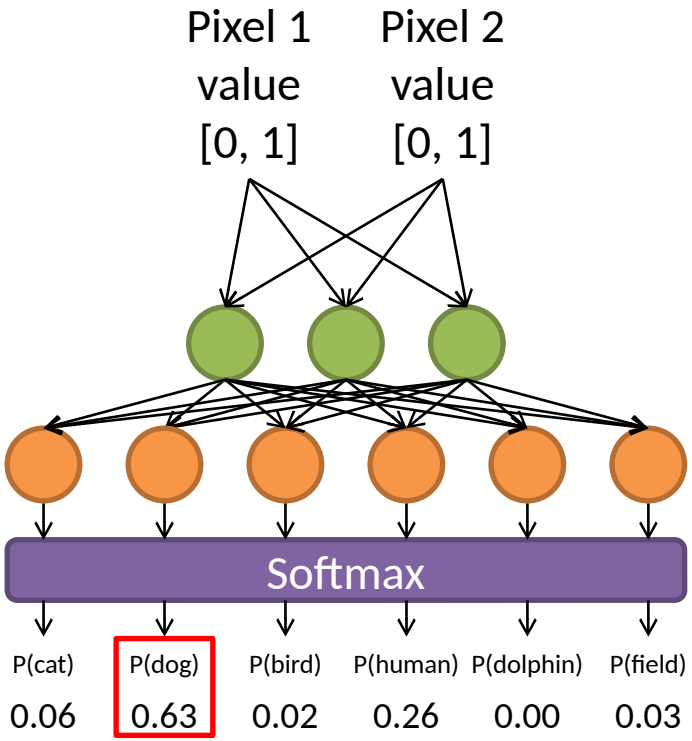
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Neural Network (NN)



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Neural Network (NN)



P(cat) + P(dog) + P(bird) + P(human) + P(dolphin) + P(field) = 1.0

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Challenge: Illumination

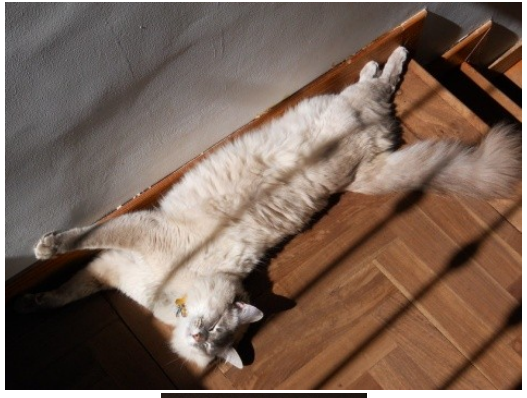




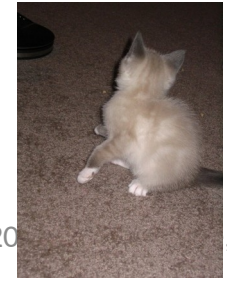
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Challenge: Deformation









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Challenge: Occlusion







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